

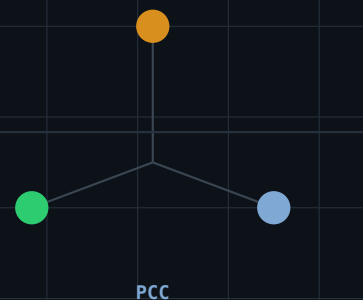
INDUSTRIAL MICROGRID CONTROLLER

Intelligent Battery Energy Storage System (BESS) Dispatch

TECHNICAL PROJECT MANUAL · REV. A

By Sipho Lucky Sibanda

Platform: Siemens S7-1500 (SCL) / CODESYS-portable Structured Text
Simulation & Portfolio Engineering Build · Not for Utility Interconnection Use



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FRONT MATTER

Document Control & Disclaimer

This document is a self-authored technical project manual produced as part of a personal engineering portfolio. It describes the design, control philosophy, and simulated validation of an industrial microgrid controller dispatching a behind-the-meter battery energy storage system, built to demonstrate demand-charge economics, time-of-use energy arbitrage, and power-quality control for automation, controls, and electrical/power systems engineering roles.

The system described here was developed and tested in simulation only (PLCSIM-style forcing of inputs and desktop review). No part of this project has been installed, commissioned, or verified on physical power systems hardware, and it must not be treated as a certified utility interconnection or protection system.

HONEST LIMITATION — Portfolio project, not a certified interconnection system

A real behind-the-meter BESS installation requires a utility interconnection study, protection coordination, and compliance with grid codes such as IEEE 1547. Figures, thresholds, and I/O in this manual are engineering-realistic but illustrative.

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Author	Sipho Lucky Sibanda
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FRONT MATTER

Executive Summary

This is the fifth deliberate discipline shift in the wider automation portfolio, moving from validated process/safety control toward behind-the-meter power systems economics - a field where the control logic's job is as much about minimising a utility bill as it is about keeping a process running. Industrial utility tariffs are frequently dominated by demand charges: a single number, the highest 15 or 30-minute average kW demand in the billing period, that can dwarf the cost of the energy actually consumed.

FB_Microgrid_BESS_Dispatch gives a behind-the-meter battery three jobs, in a fixed and deliberate priority order: protect itself (SOC limits), shave demand peaks (the single biggest lever on the bill), and only after that, arbitrage cheap off-peak energy against expensive on-peak energy. A separate, largely independent control loop corrects the site's power factor using the same battery inverter's reactive power capability - genuinely respecting the shared apparent-power ceiling a real four-quadrant inverter operates under.

The design is explicit about honesty under stress: when a demand peak exceeds what the battery can actually cover, the function block discharges everything it safely can and flags the shortfall rather than silently under-shaving and letting the bill exceed expectations unflagged. Solar is curtailed only as a genuine last resort, after the battery is full and the utility interconnection agreement doesn't permit export.

What follows documents the tariff and interconnection context, architecture, hardware assumptions, full I/O list and configuration structure, control philosophy, complete annotated code, HMI design, alarm philosophy, and the functional test procedure used to validate the logic in simulation.

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MICROGRID BESS DISPATCH

Industry Context: Demand Charges & Behind-the-Meter Storage

Most industrial and commercial electricity tariffs bill on two separate components: energy consumed (kWh) and peak demand (kW). The demand charge is calculated from the single highest average demand recorded during a short window - typically 15 or 30 minutes - anywhere in the entire billing period. A factory that runs efficiently all month but has one 20-minute spike from starting a large motor can see that single spike drive a meaningful fraction of its entire electricity bill.

Why this makes a battery genuinely valuable, not just green

A behind-the-meter battery sized correctly can "shave" that peak by discharging exactly when the factory's net demand would otherwise exceed a target ceiling - reducing the billed demand without the factory changing its actual operations at all. This is frequently the single largest economic driver for installing a BESS at an industrial site, ahead of energy arbitrage or renewable integration on their own.

Time-of-use tariffs and arbitrage

Many utilities also charge different energy rates depending on the time of day - cheaper overnight/off-peak rates, and expensive afternoon/evening on-peak rates that coincide with system-wide demand. A battery charged during off-peak hours and discharged during on-peak hours captures that price difference directly - energy arbitrage - though, as this project's dispatch priority makes explicit (Chapter 5), that opportunity is worth less than avoiding a demand charge peak.

NOTE — Interconnection agreements matter as much as the tariff

Whether a site can export excess solar back to the grid at all is governed by its utility interconnection agreement, not just technical capability. Many industrial agreements simply don't allow export - which is why this project treats grid export as conditional (``DI_NetMeteringAllowed``) rather than assumed, with battery charging and, as a last resort, solar curtailment as the fallback.

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MICROGRID BESS DISPATCH

System Architecture & Energy Flow Overview

Solar PV, a battery energy storage system, and the utility grid all connect at a common point - the Point of Common Coupling (PCC) - feeding the factory's load. The microgrid controller measures active and reactive power at the PCC and dispatches the battery's active and reactive power to manage both the bill and the site's power quality.

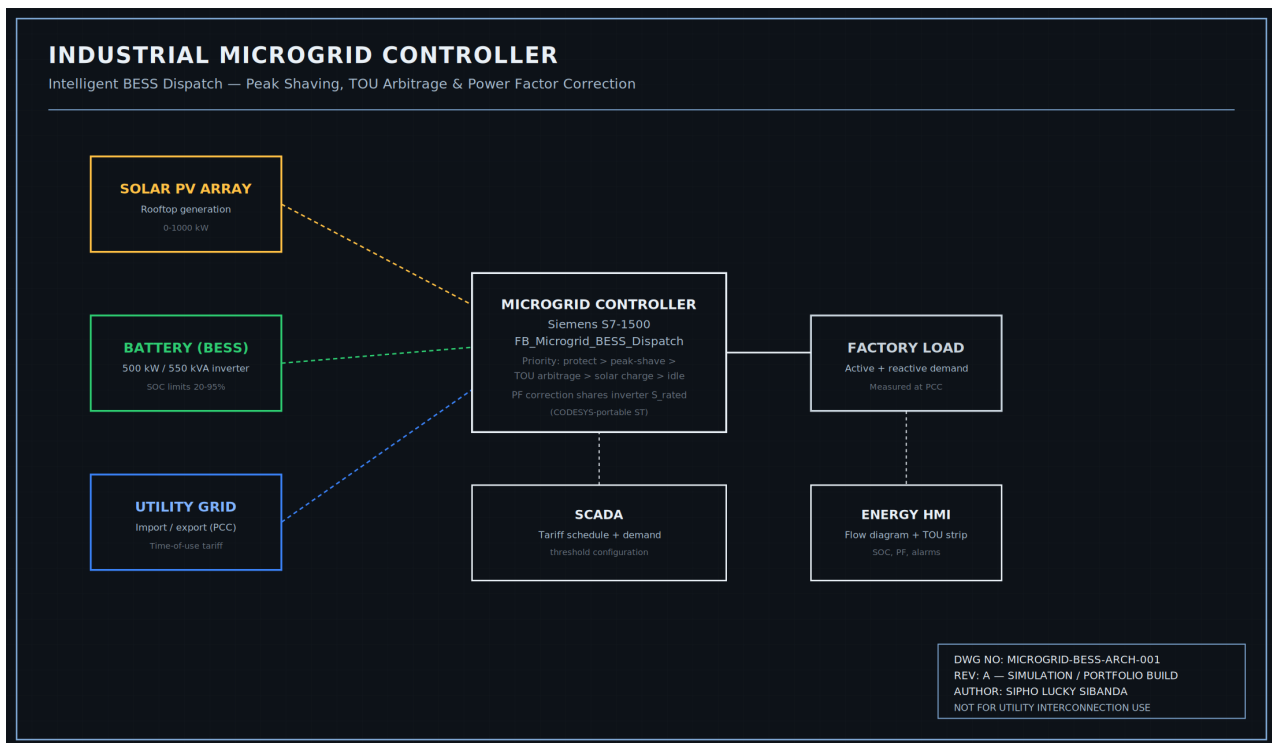


Figure 2.1 — Energy flow and control architecture. Dashed lines represent PLC signal and configuration paths; the PCC is where solar, battery, grid, and factory load all meet electrically.

Control hierarchy

- Field layer — solar PV inverter, battery energy storage system with its own BMS, main utility meter, and PCC power quality metering.
- Control layer — a single PLC function block, FB_Microgrid_BESS_Dispatch, hosted on a Siemens S7-1500 (or CODESYS-based substation/microgrid controller), running the full dispatch priority chain.
- Supervisory layer — SCADA (tariff schedule and demand threshold configuration) and the energy HMI (live flow visualisation for site operators).

Why net load, not gross load, drives the dispatch decision

The function block computes $\text{NetLoad_kW} = \text{FactoryLoad} - \text{SolarGeneration}$ as its very first step, and every dispatch decision downstream operates on that net figure. Solar generation is, in effect, treated as a form of demand reduction the battery doesn't need to duplicate - the battery only needs to cover whatever the site's own generation isn't already covering.

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MICROGRID BESS DISPATCH

Hardware & Instrumentation Specification

As with the rest of this portfolio, the logic here is platform-portable but written against a concrete reference platform so thresholds are grounded rather than arbitrary.

Component	Representative Spec	Role
Microgrid Controller CPU	Siemens SIMATIC S7-1500	Executes FB_Microgrid_BESS_Dispatch
Battery Inverter	500 kW / 550 kVA, four-quadrant	Active + reactive power dispatch
Battery Management System (BMS)	CAN/Modbus link to the controller	SOC feedback and protection
Main Utility Meter	Revenue-grade, 4-20mA or Modbus	Factory load and PCC power quality
Solar PV Inverter	Grid-tied string or central inverter	Solar generation feedback
Real-Time Clock	Controller-resident RTC	Time-of-use schedule reference

VERIFIED / DESIGN NOTE — Why the inverter's apparent power rating matters as much as its kW rating

A battery inverter rated 500 kW active power might carry a 550 kVA apparent power rating - the extra headroom exists specifically so some reactive power capability remains even while dispatching significant active power. Sizing an inverter on active power alone, without considering how much reactive power support it will also be expected to provide, is a common and costly real-world design mistake.

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MICROGRID BESS DISPATCH

I/O List & Configuration Parameters

The table below is the working I/O list for FB_Microgrid_BESS_Dispatch (see also docs/IO_List.md in the repository, and Appendix A of this manual).

Configuration parameters (downloaded from SCADA)

Field	Typical Value
DemandThreshold_kW	800 kW
OnPeak_Start_hr / OnPeak_End_hr	14.0 / 20.0
OffPeak_Start_hr / OffPeak_End_hr	22.0 / 6.0
Target_PF	0.95
Min_SOC_Pct / Max_SOC_Pct	20.0 / 95.0
Battery_Rated_kW	500 kW
Battery_Rated_kVA	550 kVA
Arbitrage_Charge_SOC_Target_Pct	90.0 %

Key inputs and outputs

Tag	Description	Signal
AI_FactoryLoad_kW / AI_SolarGeneration_kW	Site metering	4-20mA x2
AI_BatterySOC_Pct / AI_ReactivePower_kVAR	Battery + PCC feedback	Mixed x2
AI_HourOfDay_Decimal	TOU schedule reference	From RTC
DI_NetMeteringAllowed / DI_System_Enable	Interconnection + enable	Digital x2
AO_Battery_ActivePower_kW / ReactivePower_kVAR	Battery dispatch commands	4-20mA x2
AO_SolarCurtailment_Pct	Last-resort curtailment command	4-20mA
BESS_Mode / RatePeriod / Calculated_PF / SystemStatus	State & power quality data	Mixed

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MICROGRID BESS DISPATCH

Control Philosophy: Dispatch Priority & the Capability Curve

Two ideas define this project's control philosophy: a fixed economic priority order for active power, and a physically-grounded sharing rule between active and reactive power on the one inverter that has to deliver both.

5.1 The dispatch priority order, and why it's this order

Priority	Action	Why it outranks what's below it
1	Battery protection (SOC limits)	A damaged battery has no economic value at all - this is non-negotiable
2	Peak shaving	Demand charges are billed on a single peak - often the single largest bill line item
3	TOU arbitrage discharge (on-peak)	Expensive energy avoided, but smaller and more gradual value than a demand spike
4	TOU arbitrage charge (off-peak)	Prepares capacity for tomorrow's opportunities 2 and 3
5	Solar self-consumption / idle	Lowest-priority use of remaining capacity

This is not an arbitrary preference ordering - it's a rough ranking of financial impact per kWh of battery capacity spent, for a typical industrial tariff structure. A dispatch algorithm that let arbitrage compete on equal footing with peak shaving risks a battery that's mid-arbitrage-discharge exactly when a demand spike arrives with nothing left to cover it.

5.2 Honest shortfall reporting

If a demand peak exceeds what the battery can actually deliver - either because SOC is too low or the peak simply exceeds the inverter's rated power - the function block discharges everything it safely can and raises `Alarm_DemandExceeded` rather than pretending the peak was fully covered. A facilities manager relying on this system needs to know when a shave attempt fell short, not discover it as a surprise on next month's bill.

5.3 The shared inverter capability curve

A real four-quadrant battery inverter has one apparent power rating, S_{rated} , shared between active power (P) and reactive power (Q):

$$\sqrt{(P^2 + Q^2)} \leq S_{\text{rated}}$$

This project computes the reactive power headroom that remains after active power is already committed to peak shaving or arbitrage, rather than treating P and Q as two independent, unlimited outputs. During a large peak-shaving event, there may be very little reactive headroom left - and the power factor correction logic explicitly accepts that limit rather than exceeding the inverter's real rating.

VERIFIED / DESIGN NOTE — Why PF correction still runs during a peak-shave event, just with less to give

It would be simpler to disable power factor correction entirely whenever active power dispatch is high. This project deliberately doesn't do that - it computes whatever reactive headroom genuinely remains and uses all of it, flagging Alarm_PF_OutOfSpec only if that's insufficient to fully correct PF. Partial correction is still better than none.

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MICROGRID BESS DISPATCH

PLC Logic Walkthrough

This chapter walks through FB_Microgrid_BESS_Dispatch section by section. The full listing is reproduced in Appendix B.

6.1 Net load and battery headroom

```

NetLoad_kW := AI_FactoryLoad_kW - AI_SolarGeneration_kW;

IF AI_BatterySOC_Pct > Config.Min_SOC_Pct THEN
    Available_Discharge_kW := Config.Battery_Rated_kW;
ELSE
    Available_Discharge_kW := 0.0;
END_IF

IF AI_BatterySOC_Pct < Config.Max_SOC_Pct THEN
    Available_Charge_kW := Config.Battery_Rated_kW;
ELSE
    Available_Charge_kW := 0.0;
END_IF

```

Listing 6.1 — Protection limits computed before any economic decision is made.

6.2 Peak shaving, with honest shortfall reporting

```

ELSIF NetLoad_kW > Config.DemandThreshold_kW THEN
    RequiredShave_kW := NetLoad_kW - Config.DemandThreshold_kW;
    IF RequiredShave_kW <= Available_Discharge_kW THEN
        A0_Battery_ActivePower_kW := RequiredShave_kW;
        Alarm_DemandExceeded := FALSE;
    ELSE
        A0_Battery_ActivePower_kW := Available_Discharge_kW;
        Alarm_DemandExceeded := TRUE;
    END_IF
    BESS_Mode := MODE_PEAK_SHAVE;

```

Listing 6.2 — The battery discharges everything it safely can; a shortfall is flagged, not hidden.

6.3 TOU arbitrage - the literal "battery vs. grid" decision

```

ELSIF (RatePeriod = PERIOD_ONPEAK) AND (Available_Discharge_kW > 0.0) THEN
    A0_Battery_ActivePower_kW := MIN(NetLoad_kW, Available_Discharge_kW);
    BESS_Mode := MODE_ARBITRAGE_DISCHARGE;

ELSIF (RatePeriod = PERIOD_OFFPEAK) AND (AI_BatterySOC_Pct < Config.Arbitrage_Charge_SOC_Target_Pct)
    AND (Available_Charge_kW > 0.0) THEN
    A0_Battery_ActivePower_kW := -Available_Charge_kW;
    BESS_Mode := MODE_ARBITRAGE_CHARGE;

```

Listing 6.3 — This is the code the recruiter brief was built around: exactly when to draw from the battery versus buy from the grid.

6.4 Reactive power, sharing the inverter's rating

```
ActivePowerMagnitude_kW := ABS(A0_Battery_ActivePower_kW);
Available_S_kVA := SQRT(MAX(0.0, (Config.Battery_Rated_kVA * Config.Battery_Rated_kVA)
                             - (ActivePowerMagnitude_kW * ActivePowerMagnitude_kW)));

IF Calculated_PF < Config.Target_PF THEN
    RequiredQ_kVAR := AI_ReactivePower_kVAR * (1.0 - (Calculated_PF / Config.Target_PF));
    IF RequiredQ_kVAR > Available_S_kVA THEN
        A0_Battery_ReactivePower_kVAR := Available_S_kVA;
    ELSE
        A0_Battery_ReactivePower_kVAR := RequiredQ_kVAR;
    END_IF
    Alarm_PF_OutOfSpec := (A0_Battery_ReactivePower_kVAR < RequiredQ_kVAR);
END_IF
```

Listing 6.4 — Reactive headroom is derived FROM the active power already committed, not assumed unlimited.

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MICROGRID BESS DISPATCH

HMI Design & the Energy Flow Dashboard

The HMI mockup (hmi/index.html in the repository) is built around a live energy flow diagram - solar, battery, grid, and factory load meeting at a central PCC node - paired with a time-of-use schedule strip and live power-quality metrics, matching the "power flowing into a factory" dashboard this project was scoped against.

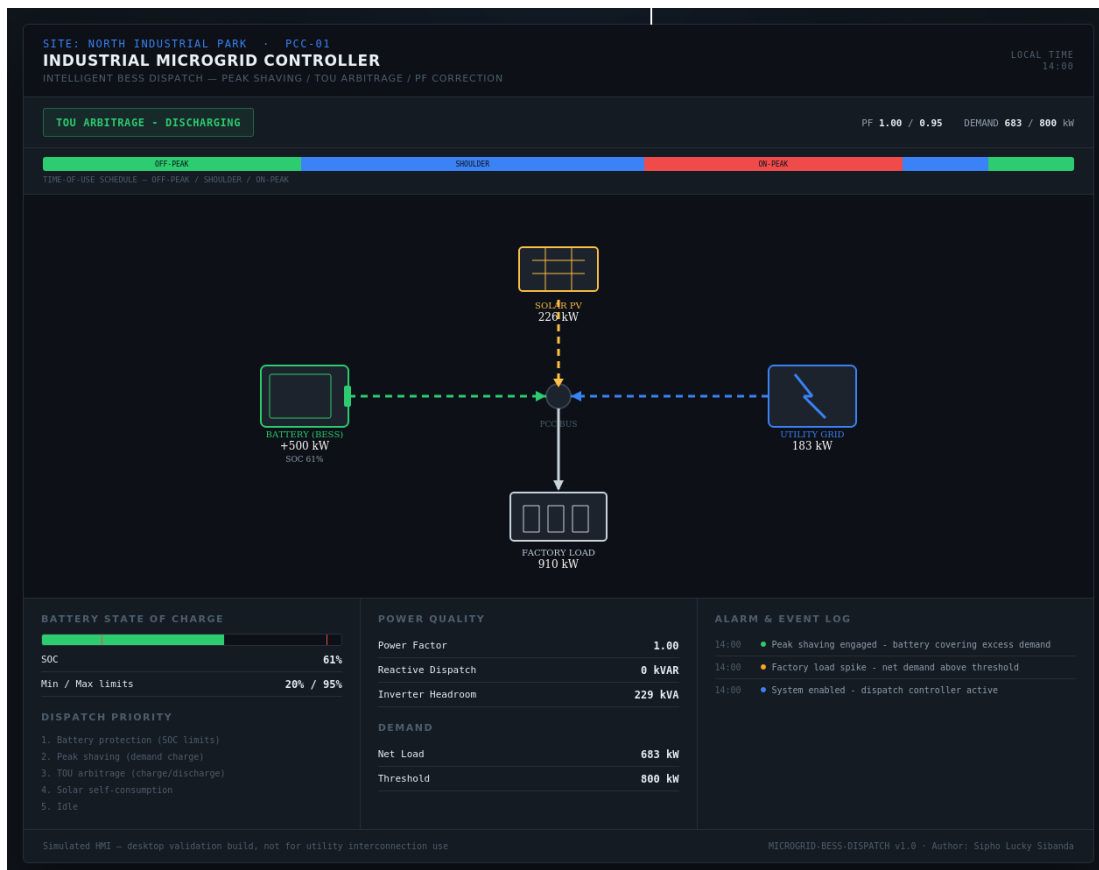


Figure 7.1 — Mid-afternoon on-peak arbitrage discharge: battery covering 500kW of the 683kW net load, animated flow lines showing direction, and the TOU strip's time marker sitting inside the on-peak (red) band.

Design decisions

- A tri-colour energy palette, not a single accent — amber for solar, green for battery, blue for grid - a sixth distinct visual identity in this portfolio, and the only one that genuinely needs three colours rather than one to represent its subject honestly.
- Animated, directional flow lines — each connection's arrowhead flips to show the actual direction of power flow (e.g. battery charging vs. discharging), tied directly to the sign convention the PLC logic itself uses.
- The TOU schedule is always visible — with a live time marker, so an operator can see at a glance not just what the system is doing now, but why, given where the site sits in its tariff cycle.

- Numbers that are internally consistent — the HMI's own dispatch simulation mirrors the same priority logic as the PLC, so battery kW + grid kW always sums to the displayed net load, the same arithmetic discipline the real function block enforces.

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MICROGRID BESS DISPATCH

Alarm Philosophy & Fail-Safe Design

Every alarm in this system maps to a specific financial or power-quality consequence, consistent with the rest of this portfolio.

Condition	Meaning	Expected Response
Alarm_SOC_Low / Alarm_SOC_High	Battery at a protective SOC limit	Informational - the battery is managing itself correctly
Alarm_DemandExceeded	Battery could not fully cover a demand peak	Review whether battery sizing or the demand threshold needs revisiting
Alarm_PF_OutOfSpec	Inverter reactive headroom insufficient for target PF	Expected during large peak-shave events; investigate if persistent outside them

Fail-safe defaults, summarised

- System disabled → battery active power commands to zero immediately, no stale dispatch commands held over.
- SOC at a protective limit → that direction (charge or discharge) is unavailable to every priority tier above idle, without exception.
- A demand peak the battery can't fully cover is reported honestly, not masked by reporting only what was achieved as if it were the whole target.

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MICROGRID BESS DISPATCH

Testing, Commissioning & FAT Procedures

The function block was validated against twelve functional test cases, anchored by a pair of tests proving the inverter capability-curve constraint is genuinely enforced rather than assumed. The full procedure is in docs/Testing_Procedures.md; the matrix is reproduced below.

#	Test Case	Expected Result
1-3	Peak shaving: engage, SOC floor, shortfall reporting	Correct discharge, correct alarm behaviour in every case
4-6	Solar excess: charge battery, curtail, export	Correct priority between charging, curtailment, and export
7-8	TOU arbitrage: on-peak discharge, off-peak charge	Battery favoured over grid on-peak; charges toward target off-peak
9	Shoulder period idle	No unnecessary cycling when there's no economic case to act
10-11	PF correction: ample headroom, constrained headroom	Full correction when possible; honest clamping and alarm when not
12	System disable	Immediate safe idle, no stale commands

ENGINEERING NOTE — Why Test 11 is the one that actually proves something

Any dispatch logic can look correct when reactive power has unlimited headroom to work with. Test 11 specifically forces a large simultaneous active power dispatch and confirms the reactive dispatch is clamped to what's genuinely left on the shared inverter rating - the difference between a plausible simulation and one that would actually respect real inverter hardware.

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MICROGRID BESS DISPATCH

Limitations, Real-World Deltas & Future Work

Naming the gap between a strong simulation and a certifiable interconnected power system is part of the engineering, consistent with every other project in this portfolio.

What a real installation would add

- Grid code compliance (e.g. IEEE 1547) — anti-islanding protection, voltage/frequency ride-through, and formal utility interconnection approval this project makes no claim to.
- Load and solar forecasting — this project reacts to current conditions; a production system would forecast the coming peak to pre-position SOC ahead of it.
- Protection coordination — real breaker and relay coordination between the BESS, solar, and utility connection, well beyond this project's dispatch-only scope.
- Multi-asset dispatch — a real site may have multiple battery units or multiple solar arrays requiring their own internal load-sharing, not modelled here.

10.1 Hazard & safeguard register (HAZOP-style)

Hazard	Cause	Safeguard in This Design
Battery degradation from deep cycling	Dispatch logic ignoring SOC limits under pressure	SOC protection is priority 1, unconditionally, before any economic decision
Inverter overload from P+Q dispatch	Active and reactive power treated as independent	Explicit shared capability-curve calculation (Chapter 5.3, 6.4)
Uncontrolled export during an outage	No anti-islanding logic modelled	Named gap - a real system requires IEEE 1547 compliant anti-islanding protection
Silent under-performance against demand charge target	A shortfall reported as if fully corrected	Alarm_DemandExceeded raised honestly whenever the battery can't fully cover a peak

Where this project could go next

- Add a simple load-forecasting model to pre-position SOC ahead of an expected peak, rather than reacting only once the peak has already begun.
- Model anti-islanding protection and a formal IEEE 1547 compliance checklist.
- Extend to a second battery asset, demonstrating fleet-level dispatch coordination.

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APPENDIX

Appendix A — I/O Quick Reference

Tag	Dir.	Type	Notes
Config	IN	ST_MicrogridConfig	From SCADA
AI_FactoryLoad_kW / SolarGeneration_kW	IN	REAL x2	Site metering
AI_BatterySOC_Pct	IN	REAL	20-95% limits
AI_ReactivePower_kVAR	IN	REAL	PCC measurement
AI_HourOfDay_Decimal	IN	REAL	TOU schedule ref
DI_NetMeteringAllowed	IN	BOOL	Interconnection agreement
DI_System_Enable	IN	BOOL	Master enable
AO_Battery_ActivePower_kW	OUT	REAL	+discharge / -charge
AO_Battery_ReactivePower_kVAR	OUT	REAL	Shares S _{rated} with above
AO_SolarCurtailment_Pct	OUT	REAL	Last resort only
BESS_Mode	OUT	ENUM	6-mode dispatch state
RatePeriod	OUT	ENUM	TOU period
Calculated_PF	OUT	REAL	Site power factor
Alarm_SOC_Low/High/DemandExceeded/ PF_OutOfSpec	OUT	BOOL x4	Process alarms
SystemStatus	OUT	STRING	HMI display

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APPENDIX

Appendix B — Full Structured Text Listing

Complete, unedited listing of src/Microgrid_BEES_Dispatch.st.

```
(*
=====
PROJECT : Industrial Microgrid Controller
          Intelligent Battery Energy Storage System (BEES) Dispatch
MODULE   : FB_Microgrid_BEES_Dispatch
PLATFORM : IEC 61131-3 Structured Text (Siemens SCL / CODESYS-portable)
AUTHOR   : Sipho Lucky Sibanda
          See Chapter 5 for the dispatch priority and capability-curve discussion.
=====
*)

TYPE E_RatePeriod : (PERIOD_OFFPEAK, PERIOD_SHOULDER, PERIOD_ONPEAK); END_TYPE

TYPE E_BEES_Mode :
(
    MODE_IDLE, MODE_PEAK_SHAVE, MODE_ARBITRAGE_DISCHARGE,
    MODE_ARBITRAGE_CHARGE, MODE_SOLAR_CHARGE, MODE_PROTECT
);
END_TYPE

TYPE ST_MicrogridConfig :
STRUCT
    DemandThreshold_kW : REAL;
    OnPeak_Start_hr : REAL; OnPeak_End_hr : REAL;
    OffPeak_Start_hr : REAL; OffPeak_End_hr : REAL;
    Target_PF : REAL;
    Min_SOC_Pct : REAL; Max_SOC_Pct : REAL;
    Battery_Rated_kW : REAL; Battery_Rated_kVA : REAL;
    Arbitrage_Charge_SOC_Target_Pct : REAL;
END_STRUCT
END_TYPE

FUNCTION_BLOCK FB_Microgrid_BEES_Dispatch
VAR_INPUT
    Config : ST_MicrogridConfig;
    AI_FactoryLoad_kW : REAL; AI_SolarGeneration_kW : REAL;
    AI_BatterySOC_Pct : REAL; AI_ReactivePower_kVAR : REAL;
    AI_HourOfDay_Decimal : REAL;
    DI_NetMeteringAllowed : BOOL; DI_System_Enable : BOOL;
END_VAR
```

```

VAR_OUTPUT
  A0_Battery_ActivePower_kW : REAL := 0.0;
  A0_Battery_ReactivePower_kVAR : REAL := 0.0;
  A0_SolarCurtailment_Pct : REAL := 0.0;
  BESS_Mode : E_BESS_Mode := MODE_IDLE;
  RatePeriod : E_RatePeriod := PERIOD_OFFPEAK;
  Calculated_PF : REAL := 1.0;
  Alarm_SOC_Low : BOOL := FALSE; Alarm_SOC_High : BOOL := FALSE;
  Alarm_DemandExceeded : BOOL := FALSE; Alarm_PF_OutOfSpec : BOOL := FALSE;
  SystemStatus : STRING[28] := 'STANDBY';
END_VAR

VAR
  NetLoad_kW : REAL;
  Available_Discharge_kW : REAL; Available_Charge_kW : REAL;
  RequiredShave_kW : REAL; GridImport_kW : REAL;
  Available_S_kVA : REAL; RequiredQ_kVAR : REAL;
  ActivePowerMagnitude_kW : REAL;
END_VAR

// 1. RATE PERIOD FROM TOU SCHEDULE
IF (AI_HourOfDay_Decimal >= Config.OnPeak_Start_hr) AND (AI_HourOfDay_Decimal <
Config.OnPeak_End_hr) THEN
  RatePeriod := PERIOD_ONPEAK;
ELSIF (AI_HourOfDay_Decimal >= Config.OffPeak_Start_hr) OR (AI_HourOfDay_Decimal <
Config.OffPeak_End_hr) THEN
  RatePeriod := PERIOD_OFFPEAK;
ELSE
  RatePeriod := PERIOD_SHOULDER;
END_IF

// 2. NET LOAD AND BATTERY HEADROOM
NetLoad_kW := AI_FactoryLoad_kW - AI_SolarGeneration_kW;
IF AI_BatterySOC_Pct > Config.Min_SOC_Pct THEN Available_Discharge_kW := Config.Battery_Rated_kW;
ELSE Available_Discharge_kW := 0.0; END_IF
IF AI_BatterySOC_Pct < Config.Max_SOC_Pct THEN Available_Charge_kW := Config.Battery_Rated_kW;
ELSE Available_Charge_kW := 0.0; END_IF
Alarm_SOC_Low := AI_BatterySOC_Pct <= Config.Min_SOC_Pct;
Alarm_SOC_High := AI_BatterySOC_Pct >= Config.Max_SOC_Pct;

```

```

// 3. ACTIVE POWER DISPATCH - FIXED PRIORITY ORDER
IF NOT DI_System_Enable THEN
    AO_Battery_ActivePower_kW := 0.0; BESS_Mode := MODE_IDLE;
ELSIF NetLoad_kW > Config.DemandThreshold_kW THEN
    RequiredShave_kW := NetLoad_kW - Config.DemandThreshold_kW;
    IF RequiredShave_kW <= Available_Discharge_kW THEN
        AO_Battery_ActivePower_kW := RequiredShave_kW; Alarm_DemandExceeded := FALSE;
    ELSE
        AO_Battery_ActivePower_kW := Available_Discharge_kW; Alarm_DemandExceeded := TRUE;
    END_IF
    BESS_Mode := MODE_PEAK_SHAVE;
ELSIF NetLoad_kW < 0.0 THEN
    Alarm_DemandExceeded := FALSE;
    IF Available_Charge_kW > 0.0 THEN
        AO_Battery_ActivePower_kW := MAX(NetLoad_kW, -Available_Charge_kW);
        BESS_Mode := MODE_SOLAR_CHARGE; AO_SolarCurtailment_Pct := 0.0;
    ELSIF NOT DI_NetMeteringAllowed THEN
        AO_Battery_ActivePower_kW := 0.0; BESS_Mode := MODE_PROTECT;
        AO_SolarCurtailment_Pct := 100.0 * (ABS(NetLoad_kW) / MAX(AI_SolarGeneration_kW, 0.1));
    ELSE
        AO_Battery_ActivePower_kW := 0.0; BESS_Mode := MODE_IDLE; AO_SolarCurtailment_Pct := 0.0;
    END_IF
ELSIF (RatePeriod = PERIOD_ONPEAK) AND (Available_Discharge_kW > 0.0) THEN
    AO_Battery_ActivePower_kW := MIN(NetLoad_kW, Available_Discharge_kW);
    BESS_Mode := MODE_ARBITRAGE_DISCHARGE; Alarm_DemandExceeded := FALSE;
ELSIF (RatePeriod = PERIOD_OFFPEAK) AND (AI_BatterySOC_Pct < Config.Arbitrage_Charge_SOC_Target_Pct)
    AND (Available_Charge_kW > 0.0) THEN
    AO_Battery_ActivePower_kW := -Available_Charge_kW;
    BESS_Mode := MODE_ARBITRAGE_CHARGE; Alarm_DemandExceeded := FALSE;
ELSE
    AO_Battery_ActivePower_kW := 0.0; BESS_Mode := MODE_IDLE; Alarm_DemandExceeded := FALSE;
END_IF
GridImport_kW := NetLoad_kW - AO_Battery_ActivePower_kW;

```

```

// 4. REACTIVE POWER / POWER FACTOR CORRECTION
ActivePowerMagnitude_kW := ABS(A0_Battery_ActivePower_kW);
Available_S_kVA := SQRT(MAX(0.0, (Config.Battery_Rated_kVA * Config.Battery_Rated_kVA)
                             - (ActivePowerMagnitude_kW * ActivePowerMagnitude_kW)));

IF ABS(AI_FactoryLoad_kW) > 0.01 THEN
    Calculated_PF := AI_FactoryLoad_kW
        / SQRT((AI_FactoryLoad_kW * AI_FactoryLoad_kW) + (AI_ReactivePower_kVAR *
AI_ReactivePower_kVAR));
ELSE
    Calculated_PF := 1.0;
END_IF

IF Calculated_PF < Config.Target_PF THEN
    RequiredQ_kVAR := AI_ReactivePower_kVAR * (1.0 - (Calculated_PF / Config.Target_PF));
    IF RequiredQ_kVAR > Available_S_kVA THEN
        A0_Battery_ReactivePower_kVAR := Available_S_kVA;
    ELSE
        A0_Battery_ReactivePower_kVAR := RequiredQ_kVAR;
    END_IF
    Alarm_PF_OutOfSpec := (A0_Battery_ReactivePower_kVAR < RequiredQ_kVAR);
ELSE
    A0_Battery_ReactivePower_kVAR := 0.0; Alarm_PF_OutOfSpec := FALSE;
END_IF

// 5. STATUS TEXT
IF NOT DI_System_Enable THEN SystemStatus := 'STANDBY';
ELSIF Alarm_DemandExceeded THEN SystemStatus := 'PEAK SHAVE - CAPACITY LIMIT';
ELSE
    CASE BESS_Mode OF
        MODE_PEAK_SHAVE: SystemStatus := 'PEAK SHAVING ACTIVE';
        MODE_ARBITRAGE_DISCHARGE: SystemStatus := 'TOU ARBITRAGE - DISCHARGING';
        MODE_ARBITRAGE_CHARGE: SystemStatus := 'TOU ARBITRAGE - CHARGING';
        MODE_SOLAR_CHARGE: SystemStatus := 'CHARGING FROM EXCESS SOLAR';
        MODE_PROTECT: SystemStatus := 'SOLAR CURTAILED - BATTERY FULL';
        ELSE SystemStatus := 'IDLE';
    END_CASE
END_IF

END_FUNCTION_BLOCK

```

Listing B.1 — Complete FB_Microgrid_BESS_Dispatch source.

OC

APPENDIX

Appendix C — Glossary

Term	Meaning
BESS	Battery Energy Storage System
Demand charge	A utility bill component based on peak kW demand, not kWh consumed
Peak shaving	Using stored energy to reduce a billed demand peak
TOU / arbitrage	Time-of-Use tariff; charging cheap and discharging expensive periods
PCC	Point of Common Coupling — where generation, storage, and the grid meet the load
Power factor (PF)	$P / \sqrt{P^2 + Q^2}$ — a measure of how efficiently power is delivered
Reactive power (kVAR)	Non-working power associated with inductive/capacitive loads
Four-quadrant inverter	An inverter able to source/sink both active and reactive power
SOC	State of Charge — a battery's remaining energy as a percentage of capacity
Net metering	A utility arrangement allowing excess generation to be exported/credited
IEEE 1547	The standard governing grid interconnection of distributed energy resources



CLOSING

About the Author

Sipho Lucky Sibanda is an automation and controls engineer building a multi-disciplinary portfolio spanning marine systems, avionics, architectural technology, applied AI, biotechnology, electrical power systems, and industrial automation. This manual documents the fifth deliberate discipline shift in that portfolio, following the same documentation standard as every other project in the series: full logic, I/O documentation, a live HMI, functional test procedures, and an honest account of what separates a strong simulation from a certifiable production system.

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End of document.